William Mundine & Leon Ellis

Predicting Points Per Game: 2023-2024 NBA Season.

Exploratory Data Analysis

One of the clearer predictive relationships is that between a player's number of games played and average points per game (`PTS`). The most valuable players will be chosen to play in more games (`G`). This logic extends to games started (`GS`). As these two predictors have a fairly strong correlation coefficient, 0.649, including both could harm our models through collinearity. We first considered using `G` alone, but its correlation with `PTS` was only moderate, 0.579. The correlation between `GS` and `PTS` was stronger, 0.804, but games started by itself would fail to capture certain cases. For example, one of the leading scorers in the dataset was Joel Embiid who had 34.7 points per game, but only played 39 games due to an injury. Of those 39 games, however, Embiid started in each one. Similarly, Ja Morant averaged 35.3 points per game, but only played in nine games, all nine of which he started in. To mitigate the impacts of these players, we instead created a new variable `starter` from dividing games started by games (i.e. the percentage of a player's games they started in.) This new variable has a correlation with `PTS` of 0.835.

Another relationship we considered was that between a player's position and their average points per game. We found that measures of central tendency and interquartile ranges were fairly similar across the positions with far greater variation within a position than between them as depicted in Figure 1-1. (Note: eight players with hybrid positions were not included in Figure 1-1 for the sake of simplicity.) For that reason, we did not include position directly in the final model.

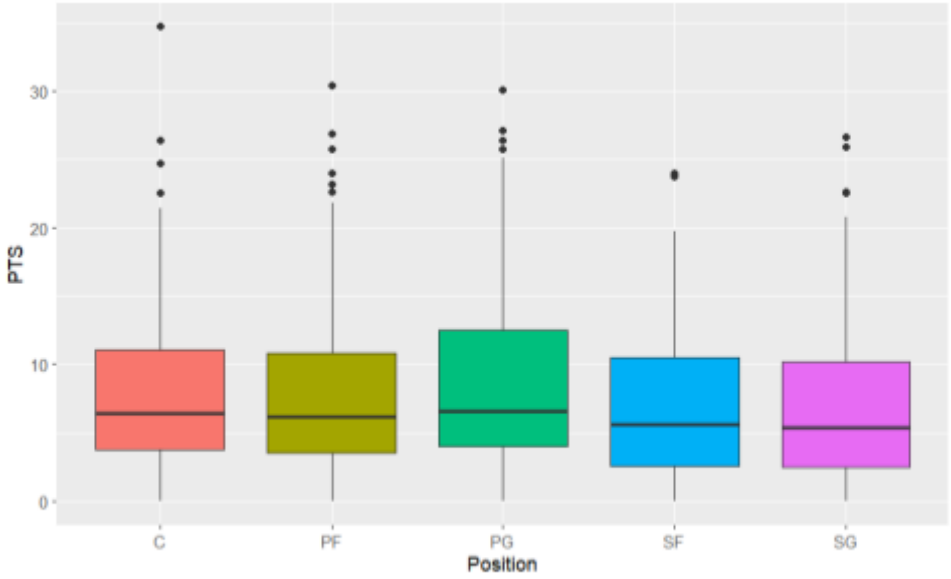


Figure 1-1

The correlations of the numeric variables with PTS are included in Figure 1-2. The correlation table indicates a number of relationships worth exploring. First is that between `TOV` (turnovers) and `PTS`. Initial assumptions might assume the two are negatively correlated since a turnover is losing the ball on offense, but the variable actually serves as a stand-in for ball possession. Ball-dominant players may turn the ball over more but also score more points. High scoring players may also have more leeway for mistakes like turnovers because of their scoring potential. Figure 1-3 is a scatterplot of turnovers and PTS.

Correlation with PTS

Predictor	Correlation	Predictor	Correlation
MP	0.890	G	0.579
TOV	0.888	BLK	0.447
RA	0.853	ORB	0.385
starter	0.835	FT.	0.369
GS	0.804	X3P.	0.318
AST	0.778	eFG.	0.304
DRB	0.770	FG.	0.286
TRB	0.702	X2P.	0.200
STL	0.686	Age	0.116
PF	0.653	Rk	-0.045

Figure 1-2

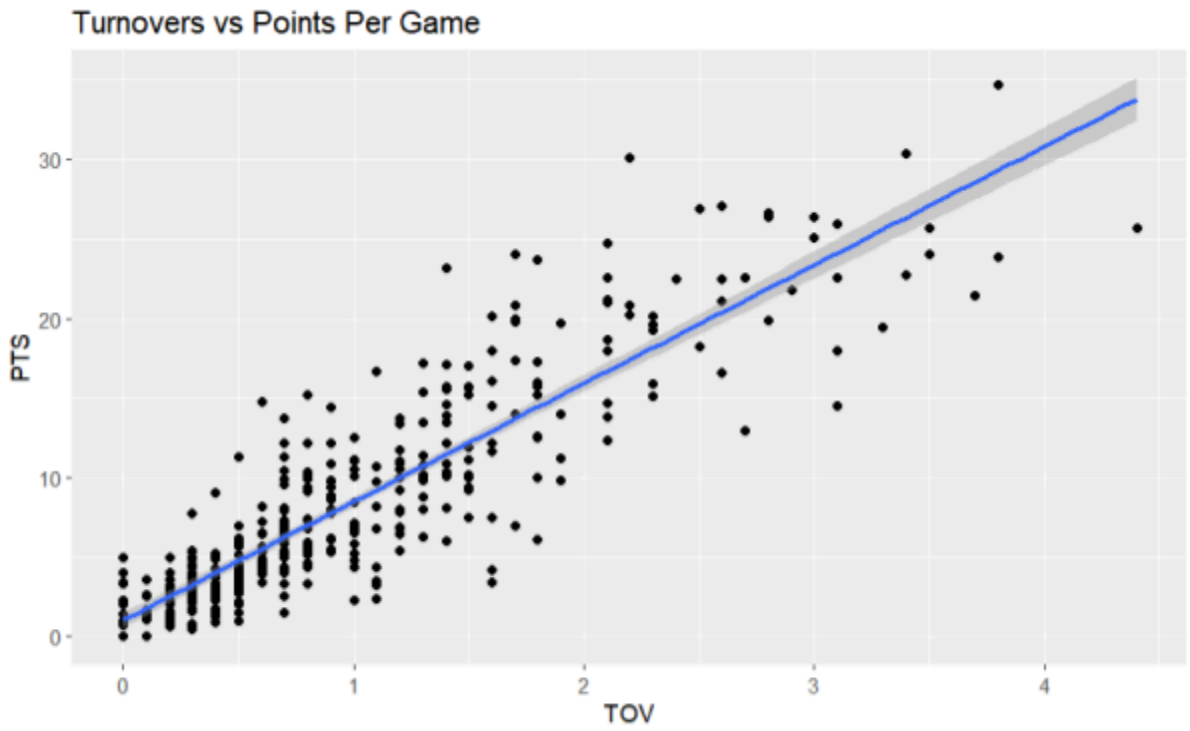


Figure 1-3

Another curious relationship is that between points per game and Rebounds and Assists (`RA`), a new variable created based on its use in sports betting contexts. It is simply the combination of total rebounds (`TRB`) and assists (`AST`). While it does have a high correlation, it is not easily interpretable. A one unit increase in `RA` having some effect is not actionable. Figure 1-4 is a scatterplot of `RA` and `PTS`.

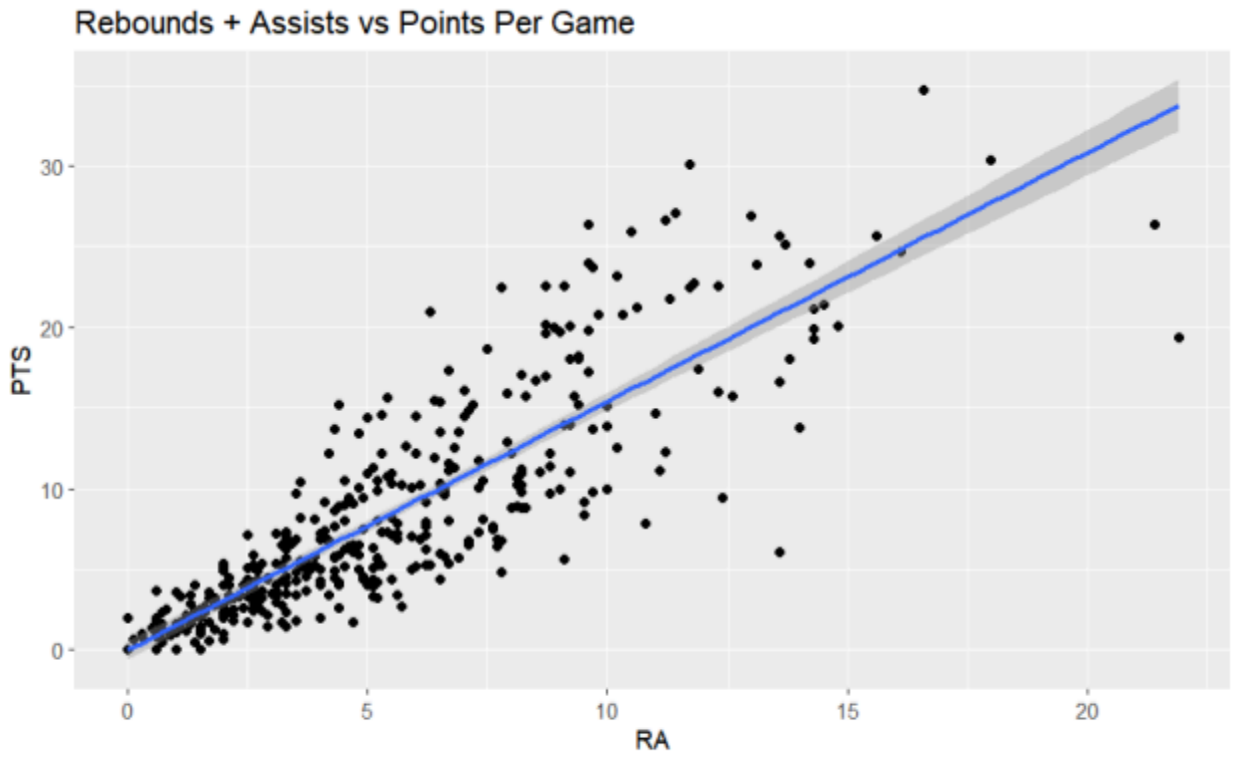


Figure 1-4

Some relationships we expected to be stronger were those between `PTS` and the shooting accuracy variables. Their densities are shown in Figure 1-5. One indicator of low correlation from the density plot is low variance. If everyone shoots at the same level of accuracy, there is no way to use it as a predictor of `PTS`. It is not a coincidence that the most spread out predictors – FT through its high variance and `X3P.` through its high density at 0% – have the highest correlations with points per game of the four in question. Effective field goal percentage was not included as it is not as easily interpreted.

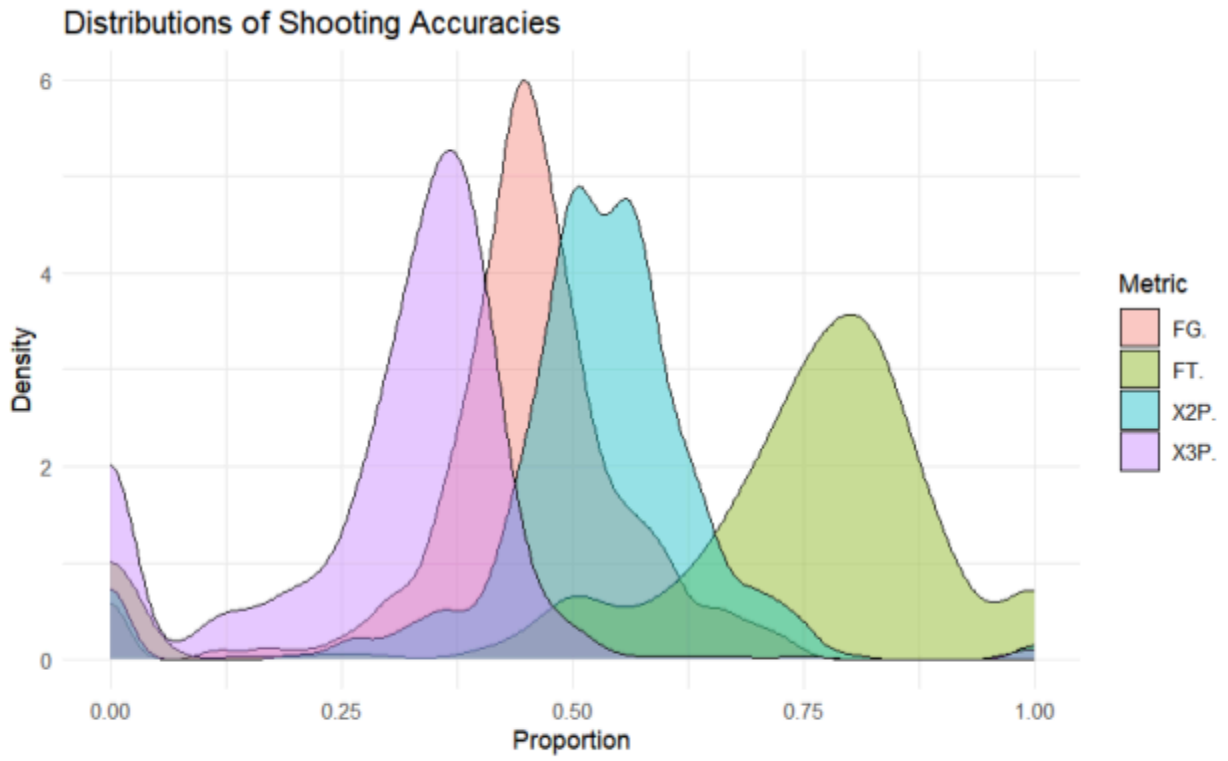


Figure 1-5

Model Building & Justification

The model built for predicting `PTS` is a multiple linear regression model estimated using Ordinary Least Square (OLS). The OLS model estimates the relationship between the predictors (`starter`, `MP`, `TOV`, `DRB`, `X3P.`, `X2P.`, and `FT.`) and the response variable (`PTS`) by minimizing the sum of squared residuals. In the model preparation `PTS` (points per game) was transformed to log(`PTS` +1) to prevent heteroscedasticity at higher fitted values and prevent negative predictions for players with low stats. This model was chosen over many other models based on many factors such as: multicollinearity, root-mean-square-error (RMSE) of the test data, p-values, and interpretability. Only one predictor had a slight multicollinearity issue based on the variance inflation factor (VIF). Minutes played (`MP`) had a VIF of 7.325 which indicates a slight multicollinearity issue since it is greater than 5. RMSE represents on average how much the model prediction deviates from the actual prediction. The RMSE of the multiple linear regression model was 2.10 meaning that, on average, the predicted `PTS` +1 differs by 2.10 from the actual `PTS`. The model is also interpretable and it is clear and meaningful.

The predictors (`starter`, `MP`, `TOV`, `DRB`, `X3P.`, `X2P.`, and `FT.`) were all used in the model because they all are believed to impact `PTS` in some way. We included these predictors based on a correlation table - see Figure 1-2. The predictors `starter`, `MP`, `TOV`, and `DRB` all

have high correlation with `PTS`; therefore, they were added to the model. Although, analysis reveals that `FT.`, `X3P.`, and `X2P.` do not have a strong correlation with points, adding the three predictors had a significant impact on the prediction of `PTS`+1.

Model Evaluation: Diagnostics

We will first evaluate our model on its adherence to OLS assumptions. In Figure 3-1, the Residuals vs Fitted plot, we see that there is very mild heteroscedasticity with a slightly tighter spread of residuals at higher fitted values but not at a degree to undermine the model. Linearity is also largely met with only a few overpredictions at the lower fitted values. This makes sense since there are players who stepped on the court without scoring any points. The model predicts an average 0.427 points per game for non-scoring players.

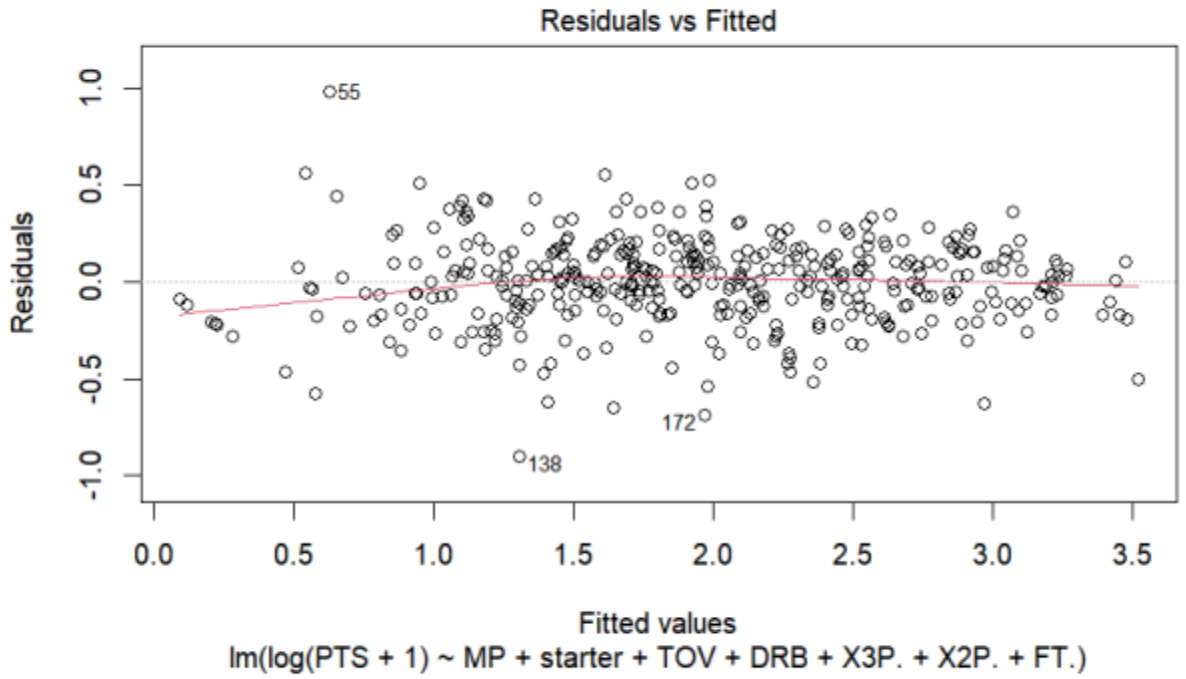


Figure 3-1

Normality is also largely met, per Figure 3-2, with a slight deviation at the lower end of theoretical quantiles where points fall below the expectation line. This deviation is not strong enough to undermine the model, however.

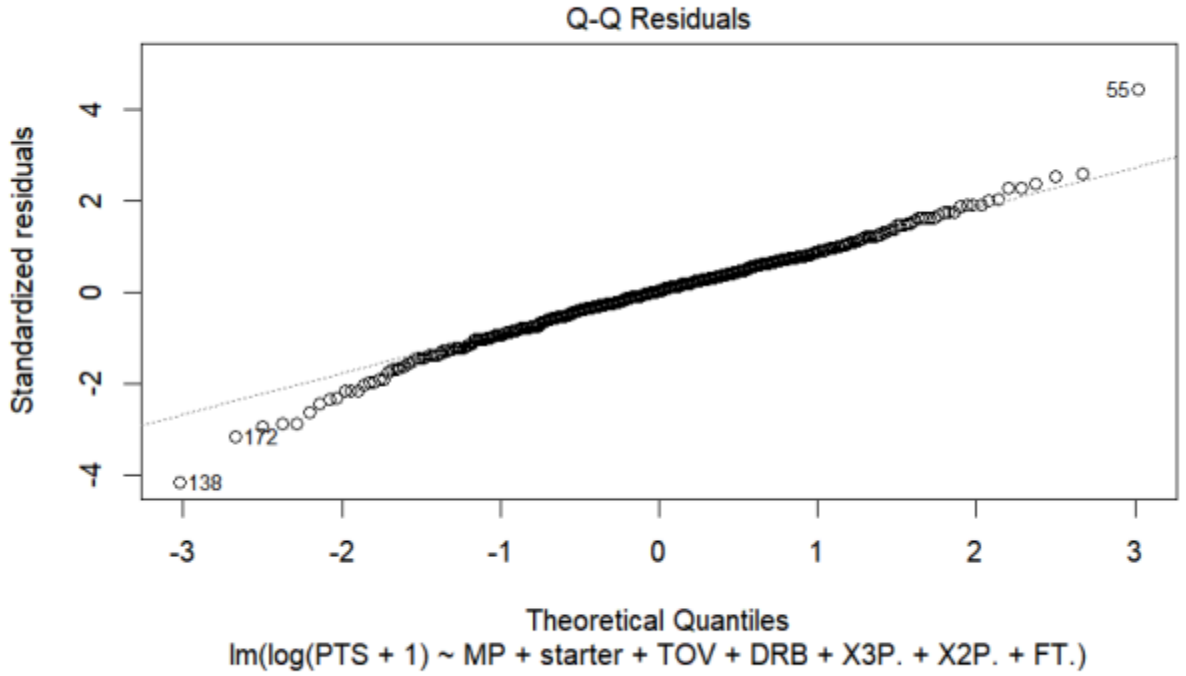


Figure 3-2

The Q-Q plot also indicates outliers at rows 55 and 138 in the dataset. These rows correspond to the players Izaiah Brockington and E.J. Liddell, respectively.

For Brockington, the Q-Q plot shows a standardized residual considerably higher than the theoretical quantile; in other words, the model underpredicts his points. In the 2023-2024 season, Brockington came off the bench for one game, played for three minutes, and scored four points. Due to his relatively low amount of time on the court to garner stats, the model uses that three minutes of play time to predict 1.10 points per game. If a model was needed for predictive accuracy at this end of the curve in particular, the number of games would likely be controlled in order to diminish the impact of cases like Brockington's.

Liddell has our lowest standardized residual. He did not start any of his eight games but played an average 2.9 minutes per game and averaged 0.5 points per game. He does, however, have a 100% two-point field goal percentage (`X2P.`) and 100% free throw percentage (`FT.`) leading to a prediction of 2.342. As with Brockington, limited data restricted the usefulness of predictions.

Figure 3-3 shows that there are no high influence points since all points are contained within the Cook's distance for the data.

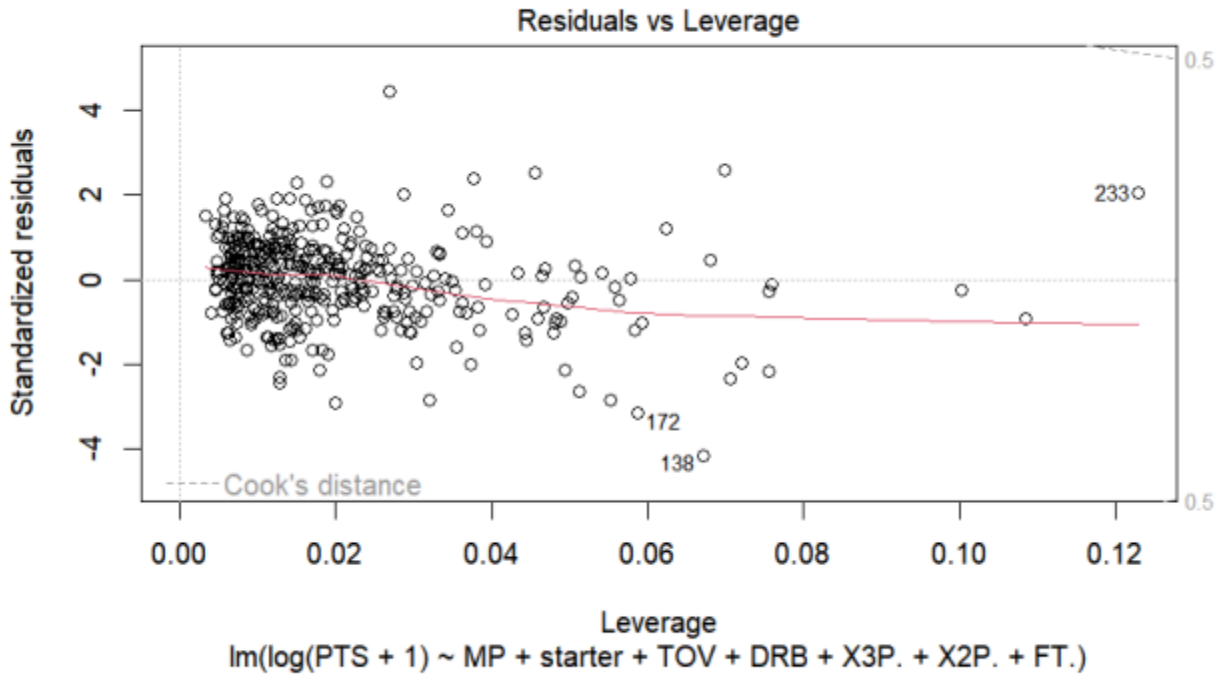


Figure 3-3

Model Evaluation: Comparison

A model we considered instead of the OLS model was a Generalized Additive Model with predictors `MP`, `TOV`, `starter`, `FT.`, `X3P.`, and `X2P`. The predictor `TOV` was unchanged due to its linearity, as seen in Figure 1-3, but the five other predictors had a spline function applied to them in order to account for non-linearity. Using cross validation with an 80-20 split, the GAM actually outperformed the final model when looking at Mean Squared Error of the test data. The split using seed one in R results in a Test MSE of 5.343 for the GAM and a Test MSE of 8.225 for the OLS model. This outperformance extended to the scoring on the Kaggle competition, MSE of the original test data.

The GAM is not without its limitations, however. The major contrast with OLS is interpretability. `TOV` remains easily interpretable as a parametric coefficient, but the other predictors must be plotted to see their relationships with `PTS`.

Like OLS, GAMs can be evaluated on assumptions like homoscedasticity. Figure 3-4 shows the plot of residuals vs. fitted values for the GAM. The funnel shape brings into question inferences drawn from the model and significance testing.

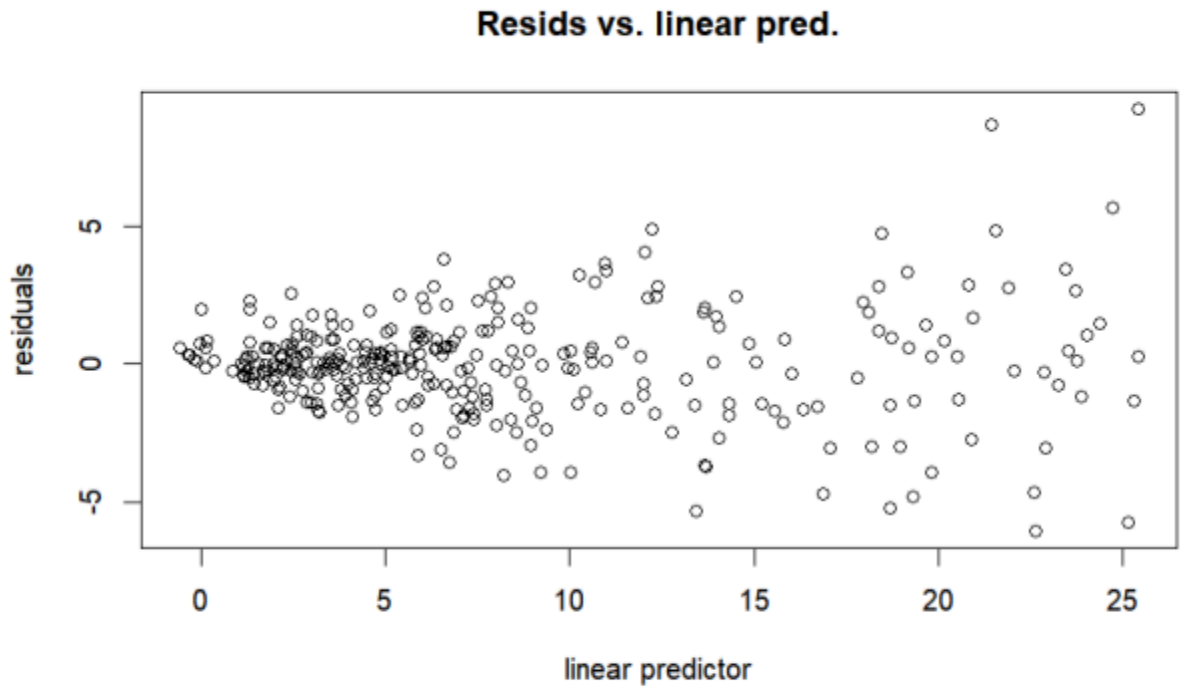


Figure 3-4

We also evaluated both models using the Akaike Information Criterion (AIC). While AIC cannot be interpreted on its own, it can be used to compare models with those with AICs closer to zero having a better trade-off between goodness of fit and model complexity. The AIC for the GAM is 1350 whereas the AIC for the OLS model is -51. The OLS model is the clear winner.

Another significant issue is the potential for negative predictions since there is no logarithmic transformation. One player with one minute on the court and no other statistics was predicted to have a -0.554 points per game. While this is the only negative prediction in the test data, without a principled approach to limiting negative predictions, there is a real risk of future data not truly capturing the underlying relationship.

Interpretation & Communication

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.064413   0.050630   1.272 0.204042
starter     -0.215914   0.061749  -3.497 0.000525 ***
MP           0.053814   0.003065  17.559  < 2e-16 ***
TOV          0.196117   0.025894   7.574 2.63e-13 ***
DRB          0.026809   0.010736   2.497 0.012932 *
X3P.         0.503782   0.083949   6.001 4.46e-09 ***
X2P.         0.751727   0.085399   8.803  < 2e-16 ***
FT.          0.259778   0.052729   4.927 1.24e-06 ***
---
```

In our OLS model there were seven predictors used for predicting points per game in an NBA season. The seven predictors were `starter`, `MP`, `TOV`, `DRB`, `X3P.`, `X2P.`, and `FT.`. Each of the predictors are statistically significant at the 0.05 level meaning there is evidence of a relationship between it and the response variable. Recall that `PTS` (points per game) was transformed into log (`PTS`+1) to accommodate for certain values. In order to interpret the predictor coefficients, the coefficients are in need of exponentiation to find the overall impact on `PTS`+1. This is represented by exp (β * unit) -1, which also demonstrates the average change in `PTS`+1 in a percentage.

Starter Percentage (`starter`):

Since `starter` is a proportion, using a 1.0 unit increase would not be reasonable. Instead a 0.1 unit increase is used. Holding all other predictors constant, a 10% increase in the ratio of games started to games played is associated with a 2.136% decrease in `PTS`. While a negative relationship seems counterintuitive, the key to interpretation is "holding all other predictors constant." From Figure 1-2, we already know that there is a positive relationship between `starter` and `PTS`, but the question being asked here is whether two players with the same stats will have the same number of `PTS` if one is a starter and the other is not. The answer lies in the fact that bench players and starters have different roles. A starter may have a number of jobs: defending, screening, passing, communication, etc. They have greater opportunity to garner stats. A bench player is often meant to focus on scoring. If they garner the same stats, that likely means they are a more efficient player, thus have higher points per game.

Minutes Played (`MP`):

Holding all other predictors constant, every 1 minute increase in `MP` there is an associated 5.529% increase in `PTS` on average. This indicates that the more minutes a player plays, the more points they will score. This is consistent in an actual NBA basketball game because the

more time the player is on the court the more opportunities they will have to score points indicating a positive relationship.

Turnovers (TOV):

Using a 1 unit increase in TOV is too large and not ideal, because of how small most of the data lies; instead, using a 0.5 unit increase is more interpretable in terms of turnovers per game. For every 0.5 turnover per game increase in TOV , on average there is an associated 10.303% increase in PTS holding all other predictors constant. This suggests that players who average more turnovers per game are the more points per game they will average. This essentially is expected with players in the NBA. Intuitively, the more ball-dominant a player is – typically players who score a significant amount of points – the more likely they are to commit turnovers. This aligns with the more turnovers per game a player has the more points per game the player scores which is a positive relationship.

Defensive Rebounds (DRB):

Holding all other predictors constant, for every 1 additional DRB there is an associated 2.717% increase in PTS on average. A positive linear relationship is plausible between DRB and PTS. In an NBA season players who average more defensive rebounds tend to play more minutes thus leading to more points per game.

3 Point Percentage (X3P.):

Instead of using a 1 unit increase in X3P., which would mean jumping from 0% to 100% X3P., a 0.1 unit increase is used. For every 0.1 field goal percentage increase in X3P., on average there is an associated 5.167% increase in PTS holding all other predictors constant. Conceptually, in order for X3P. to be significantly effective, shot attempts matter. Assuming two players have the same X3P. the player with more attempts will average more points. This suggests that in a real NBA season, X3P. has a real-world impact on PTS determined by the player's shot attempts.

2 Point Percentage (X2P.):

Similar to X3P. instead of using a 1 unit increase in X2P.a 0.1 unit increase is used. For every 0.1 field goal percentage increase in X2P., on average there is an associated 7.807% increase in PTS holding all other predictors constant. Shot attempts also matter for X2P. to be effective on PTS when compared to a realistic NBA season. Additionally, comparing two players with the same X2P. but different shot attempts, the player with more shot attempts will average more points per game. This suggests that X2P. has a practical impact on PTS determined by the amount of two-point shot attempts.

Free Throw Percentage (`FT.`):

Applying the same reasoning for `FT.`, a 0.1 unit increase in field goal percentage is used over a 1 unit increase in field goal percentage. For every 0.1 field goal percentage increase in `FT.`, on average there is an associated 2.632% increase in `PTS` holding all other predictors constant. This can be expected; however, it can also be misleading. Naturally the effect `FT.` has on points per game heavily relies on the amount of free throw attempts the player shoots. High free throw attempts indicate a huge impact on `PTS`. Low free throw attempts indicate a low impact on `PTS`. For example if a player has a `FT.` of 0.8 (80%) and only shoots 10 free throws throughout their season, there is not a great relationship with it increasing points per game. However, if a player has a `FT.` of 0.8 (80%) and shoots 100 free throws throughout the season, there will certainly be a greater impact on points per game in comparison to the player who only shot ten.